

Thematic review series: the pathogenesis of atherosclerosis. An interpretive history of the cholesterol controversy, part V: the discovery of the s...

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The first four reviews in this series (Steinberg, D. 2004. J. Lipid Res. 45: 1583-1593; Steinberg, D. 2005. J. Lipid Res. 46: 179-190; Steinberg, D. 2005. J. Lipid Res. 46: 2037-2051; Steinberg, D. 2006. J. Lipid Res. 47: 1-14) traced the gradual accumulation of evidence, evidence of several different kinds, supporting the lipid hypothesis. They tracked the history from Anitschkow's 1913 classic work on the cholesterol-fed rabbit model to the breakthrough 1984 Coronary Primary Prevention Trial, the first large, randomized, double-blind primary intervention trial showing that decreasing blood cholesterol (using cholestyramine) significantly reduces coronary heart disease events. At that point, for the first time, decreasing blood cholesterol levels became an official national public health goal. Still, only a small fraction of patients at high risk were getting appropriate cholesterol-lowering treatment, and a number of important clinical questions remained unanswered. This final review in the series traces the early studies that led to the discovery of the statins and briefly reviews the now familiar large-scale clinical trials demonstrating their safety and their remarkable effectiveness in reducing coronary heart disease morbidity and mortality.